TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Detailed Practical Study of Mixing, Blow Room, Carding and Draw Frame of Cotton Spinning

The opening line of the cotton spinning mill, studied machine by machine: how a mixing is built from bales, how the blow room opens and cleans, how the card individualises the fibres, and how the draw frame doubles and drafts the sliver straight — with the settings, observations and production figures your lab report must carry.

TE-2102 · SPINNING TECHNOLOGY-I SESSIONAL · STUDY 1 · LEVEL 2 TERM 1

1. Object and method of the study

Object: to connect the theory of TE-2101 (Spinning Technology-I) with the first four stages of the cotton line — mixing and bale management, the blow room, the carding machine and the draw frame — by identifying every working part, reading the settings as the mill has set them, and recording the production each setting gives.

Method — the same routine at every stage.

1. **Trace the material path** from bale to delivered sliver can, naming every element the material touches.
2. **Identify the working parts** listed in each section and state in one line what each does.
3. **Read and record the settings** — speeds, drafts and gauge distances. Settings are observed, never altered; ask the instructor before touching anything.
4. **Record production points** — feed weight, delivery weight or speed, waste percentage and efficiency, as in the worked checks below.
5. **Sketch** the material path (Figures 1–3 are your models) and note faults seen in operation with their likely causes from TE-2101.

2. Mixing and bale management

2.1 What to identify and observe

- **The laydown (mixing)** — count the bales opened for one mixing (commonly 30–50 bales of one mixing plan). Note that tufts are taken from *every* bale, a little at a time, so the mixing is consumed evenly over days, not bale by bale.
- **Bale identity** — read the bale marks: origin/growth, grade, and the fibre data the mill tests on receipt (staple length, fineness/micronaire, strength, trash %). Record how bales of different test values are distributed through the laydown so that no single bale dominates.
- **Conditioning** — where bales are opened and left to regain the mill's atmospheric condition before processing; note the department's temperature and relative humidity if a hygrometer is fitted.
- **Feed to the line** — the hopper feeder / bale plucker that takes small tufts from the top of the laydown; record the tuft size impression (small tufts open and clean better).

> **WHY MIX AT ALL — THE POINT TO STATE IN THE REPORT**
>
> No two bales are identical. Spinning from one bale at a time would make yarn properties jump at every bale change. A mixing spreads many bales through every metre of yarn, so length, fineness and strength are averaged, and the count and quality the mill promises stay steady from the first day of the mixing to the last.

3. The blow room

3.1 The line and its parts

Walk the line in order and identify each machine's job. A typical sequence (the lab line may differ slightly — record the actual one):

- **Bale opener / hopper feeder** — takes tufts from the laydown, first rough opening over a spiked lattice.
- **Step cleaner / porcupine opener** — beater with grid bars beneath; the first real cleaning point, where heavy trash falls through the grids.
- **Condenser and fan transport** — material travels between machines by air; identify the fan and the perforated cage (condenser) that strips the air at each drop.
- **Two-roller (scutcher) finisher** — final beater, feed rollers and lap former: the material is beaten once more, then wound into a **lap** of even weight on the lap rod between the cage rollers and calender rollers.
- **Waste collection** — find where the droppings from each beater are collected; these are weighed when cleaning efficiency is tested.

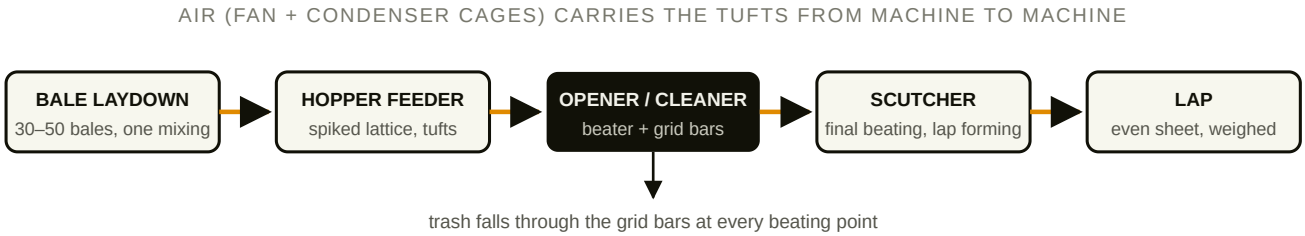


Figure 1 — The blow room sequence. Opening and cleaning alternate: every beater opens the tufts smaller and shakes trash out through its grid bars, until the scutcher delivers the material as a uniform lap for the card.

3.2 Settings and observations to record

ITEM	WHAT TO WRITE IN THE REPORT
Beater types and speeds	Blade, porcupine or pinned beater at each point; speed in rpm from the machine plate or instructor
Grid-bar setting	Inclination/opening of the grid bars under the main beater — wider opening drops more trash *and* more good fibre
Feed regulation	How even feeding is maintained (pedal regulation / photocell on modern lines) and the lap-weight control
Lap weight	Weigh a 1-yard length of lap from the scutcher; record g/m or oz/yd and compare repeated lengths for evenness
Cleaning efficiency	From the worked check below, using trash % in the mixing cotton and the waste actually extracted

WORKED CHECK — BLOW ROOM CLEANING EFFICIENCY

Cleaning efficiency = (trash removed ÷ trash fed) × 100. If the mixing cotton carries 3.0% trash and the blow room droppings amount to 2.55% of the feed weight (taken as removed trash), cleaning efficiency = 2.55 ÷ 3.0 × 100 = **85%**. The remaining trash passes forward — the card must remove most of what is left. State also the caution: waste % above the trash % means good fibre is being lost with the trash.

4. The carding machine

4.1 Parts to identify

- **Lap rod and lap** — the blow room lap unwinding at the back of the card.
- **Feed plate and feed roller** — grip the lap edge and advance it slowly and evenly to the licker-in.

- **Licker-in (taker-in)** — the saw-tooth roller that tears the lap fringe into small tufts; beneath it find the **mote knives** and **grid bars**, where most of the card's trash is thrown out.
- **Cylinder** — the large wire-clothed drum that carries the fibres forward; its undercasing controls the air flow.
- **Revolving flats** — the endless chain of clothed flats moving slowly over the top of the cylinder; the cylinder–flat working zone is where fibres are individualised. Find the flat stripping brush and where flat strips (waste) are taken off.
- **Doffer** — the smaller clothed drum that condenses the fibre film from the cylinder into a web; the **doffer comb** (or stripping roller/brush on newer machines) that strips the web off.
- **Calender rollers, trumpet and coiler** — condense the web into sliver and coil it into the delivery can.

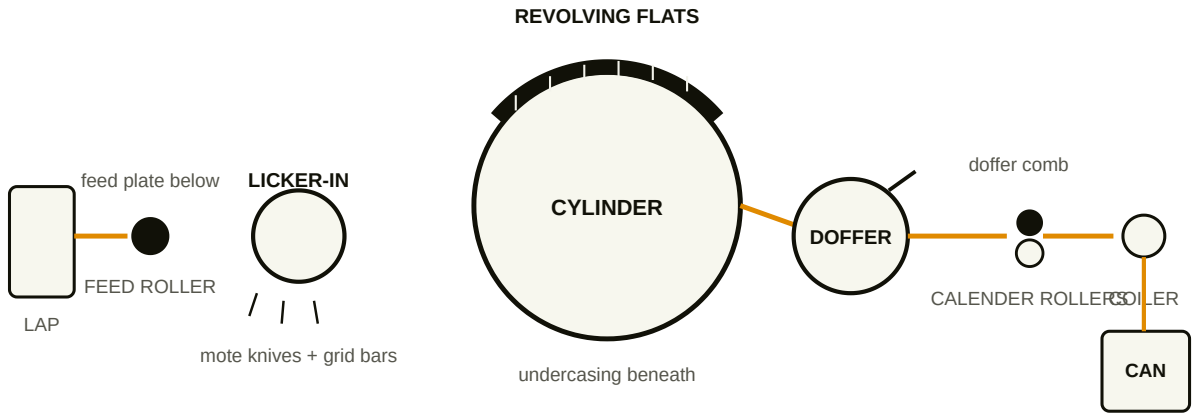


Figure 2 — Cross-section of the carding machine. The licker-in opens and cleans, the cylinder–flat zone individualises the fibres fibre-from-fibre, and the doffer condenses the film into a web that becomes sliver.

4.2 Settings and production points to record

ITEM	WHAT TO WRITE IN THE REPORT
Principal settings	Feed plate to licker-in, licker-in to cylinder, cylinder to flats (in thousandths of an inch or mm, as gauged on the machine), cylinder to doffer
Speeds	Cylinder rpm and diameter; doffer rpm; flat speed (mm/min) and its direction relative to cylinder rotation at the working face
Wastes	Licker-in droppings %, flat strips %, and total card waste % where the lab measures them

Delivered sliver	Sliver weight (g/m) by weighing a measured length; the card draft computed from lap weight and sliver weight

WORKED CHECK — CARD SPEEDS, DRAFT AND PRODUCTION

Cylinder surface speed = π × diameter × rpm = π × 1.29 m × 500 rpm ≈ **2,026 m/min** — the fastest working surface in the mill, which is why the cylinder–flat action is so intensive.
Card draft = lap weight ÷ sliver weight = 480 g/m ÷ 5.8 g/m ≈ **83**.
Doffer surface speed = π × 0.686 m × 40 rpm ≈ 86.2 m/min, which is the web (and therefore sliver) delivery speed. Production = 86.2 m/min × 60 × 5.8 g/m × 0.85 efficiency ÷ 1000 ≈ **25.5 kg/h**. Verify each figure against the lab machine's own plate values during the study.

5. The draw frame

5.1 Parts to identify

- **Creel and sliver guides** — usually 6 or 8 card sliver cans feeding one delivery; find the guide rollers and the stop motion that halts the machine when a sliver breaks or a can runs empty.
- **Drafting rollers** — bottom steel fluted rollers and top rollers (cots) in 3-over-3, 4-over-4 or 5-over-5 arrangement on the lab machine; identify back, middle and front pairs and the clearer covers/rollers that keep the cots clean.
- **Roller weighting** — dead weights, spring or pneumatic loading on the top rollers.
- **Web condenser (trumpet/funnel)** — gathers the drafted web before the calender rollers.
- **Calender rollers and coiler** — deliver the sliver and coil it into the delivery can in overlapping rings.
- **Auto-leveller (if fitted)** — find the sensing rollers at feed or the measuring trumpet at delivery and the servo that varies the draft; note whether the lab machine is open-loop or closed-loop.

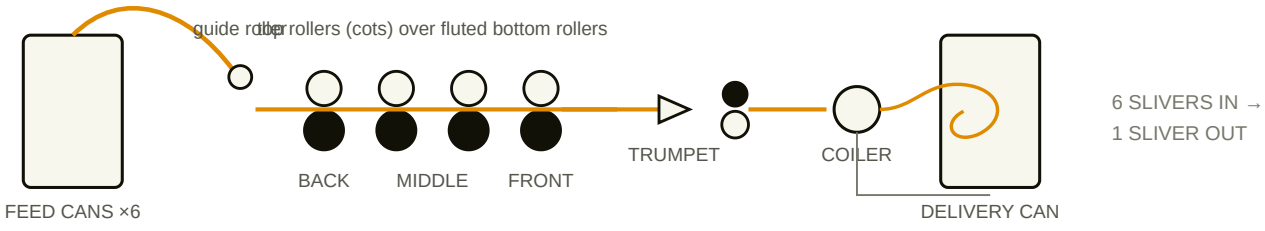


Figure 3 — The draw frame drafting arrangement. Six slivers pass together through the roller pairs; the front pair runs fastest, drafting the combined strand back to about the weight of one feed sliver.

5.2 Settings and production points to record

- **Doublings and total draft** — number of feed cans per delivery and the total draft from the change wheels; split into break draft and main draft, and state which zone carries the main draft.
- **Roller settings (gauges)** — the distances between roller pairs, set against the staple length of the mixing; note that front-zone setting is the closest.
- **Delivered sliver weight** — weigh a measured length; confirm it matches the feed sliver weight when draft = doublings.
- **Delivery speed and production** — front-roller delivery (m/min) from the machine display; production per delivery (kg/h) = delivery × 60 × sliver weight (g/m) ÷ 1000 × efficiency.

WORKED CHECK — WHY THE DELIVERED SLIVER WEIGHS THE SAME AS THE FEED

Six cans of 5.0 g/m card sliver are doubled and drafted with a total draft of 6. Combined feed = 6 × 5.0 = 30.0 g/m; after a draft of 6 the delivered sliver = 30.0 ÷ 6 = **5.0 g/m** — unchanged in weight, but the irregularities of the six slivers have been averaged together and the fibres straightened. In practice the draft is set slightly above or below the doublings to trim the delivered hank exactly.

6. What the lab report must contain

SECTION	CONTENT
Mixing record	Bale count, marks and test values in the laydown; how the mixing is consumed
Material path	Your traced path through all four stages with labelled sketches (Figures 1–3 as models)
Parts tables	Each working part, its location, and its function in one line
Settings record	The settings tables of Sections 3–5 filled with observed values — never copied from the theory sheet
Production checks	Cleaning efficiency, card production, and the draft checks, worked from your own observed figures
Faults observed	Any fault seen in operation, its probable cause from TE-2101, and the remedy the theory suggests

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why must a mixing be drawn from many bales at once instead of spinning bale by bale?
2. At the card, which zone removes most of the trash, and which zone does the fine individualising of fibres? Name the parts responsible in each.
3. A card is fed with lap of 480 g/m and delivers sliver of 5.8 g/m. What is the card draft?
4. A draw frame combines 6 slivers and applies a total draft of 6. What happens to the delivered sliver weight, and what two improvements does the passage give the sliver?
5. Which card settings most directly control nep formation, and why does a closer cylinder–flat setting reduce neps?
6. Using the worked check figures, state the card's hourly production and show the calculation from doffer speed.

Answer key

1. Because bales differ in length, fineness, strength and trash. Drawing a little from many bales at once averages those properties into every metre of yarn, so quality and count stay steady instead of jumping each time a new bale starts.
2. Most trash is removed at the **licker-in zone** — the saw-tooth roller throws the opened tufts against the mote knives and grid bars, and the great majority of the card's cleaning happens there. The fine individualising is done in the **cylinder–flat zone**, where the two clothed surfaces working close together tease the tufts apart fibre from fibre.
3. Draft = feed weight ÷ delivery weight = 480 ÷ 5.8 ≈ **83**.
4. The delivered sliver has the **same weight** as one feed sliver (6 × feed ÷ draft 6). The two improvements: **evenness** — thick and thin places of the six slivers fall at random and average out (doubling), and **fibre straightening/parallelisation** — roller drafting pulls hooked fibres straight, preparing the sliver for the finer drafting to come.
5. The **cylinder–flat setting** first (the closer the working distance, the more completely tufts are opened into single fibres, so entangled nep knots are undone instead of carried through), supported by the feed plate–licker-in and licker-in–cylinder settings, which decide how well the material is presented. A closer cylinder–flat setting also raises flat-strip waste, so it is a balance, not a free gain.
6. Doffer surface speed = π × 0.686 m × 40 rpm ≈ 86.2 m/min. Production = 86.2 × 60 min × 5.8 g/m × 0.85 ÷ 1000 ≈ **25.5 kg/h**.
